

As trained physical chemist (PhD in 1994), Siewert-Jan Marrink specialized in the use of computational modelling. Central aims of his research are: i) to unravel the lateral organization principles of biological membranes, and ii) to develop efficient computational models to simulate molecular processes. Marrink's group is well known for the development of the Martini model, a coarse-grained force field for biomolecular simulations, and has contributed fundamental understanding of membrane related processes such as fusion, domain formation, and membrane protein activity ('simulation is believing'). Marrink is also credited with several methodological advances to improve the computational microscope, including development of pioneering multi-scale simulation algorithms and high-throughput tools.

Marrink is one of the recognized world leaders and pioneer in the field of coarse-grain (CG) simulations. The Martini model developed by him is the most popular CG force field available, with thousands of users around the globe. Marrink has published >350 peerreviewed papers that received >66,000 citations (Google Scholar); He received various (personal) grants from NWO and the EU, including three NWO TOP (2004, 2008, 2014) and two ERC Advanced grants (2015, 2022). He is former chair of the board of the GBB institute, director and founder of the Berendsen Centre for Multi-scale Modeling and Material Design, is an elected Member of the Royal Netherlands Academy of Arts and Sciences, and has been granted an honorary doctorate of the Polytechnic University Bucharest (Romania). He serves on the editorial board of Biophysical Journal, and is on the advisory board of a variety of computation-oriented journals.